

Java Exception Handling Lab Report

Program: ClassCastException

1. Aim

To demonstrate the handling of ClassCastException in Java using a try-catch block when an object is cast to an incompatible data type.

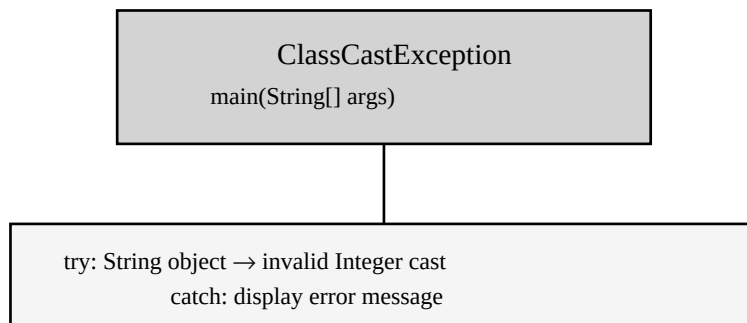
2. Description

ClassCastException is a runtime exception that occurs when an object is explicitly cast to a type that it is not compatible with. In this program, a String object containing the value "Java" is stored in an Object reference and then incorrectly cast to Integer. The exception is caught and an appropriate message is displayed.

3. Tools Used

- Java Development Kit (JDK)
- Java compiler (javac)
- IDE or command prompt
- Operating system

4. UML / Program Structure



5. Algorithm

1. Start the program.
2. Create an Object reference and assign a String value to it.
3. Attempt to cast the Object reference to Integer inside the try block.
4. Since the object is a String and not an Integer, ClassCastException occurs.
5. Catch the exception using the catch block.
6. Display the message "Invalid type casting".
7. Stop the program.

6. Complete Java Source Code

```
package exceptions;

public class Classcastexception {
    public static void main(String[] args) {
        try {
            Object obj = "Java";
```

```
        Integer n = (Integer) obj;
    } catch (ClassCastException e) {
        System.out.println("Invalid type casting");
    }
}
```

7. Sample Output

```
Invalid type casting
```

8. Result

The program successfully demonstrated ClassCastException handling. The invalid type casting was caught by the catch block, and the error message was printed without terminating the program unexpectedly.